

Market Sonata

How a year of NVIDIA stock becomes a piano sonata: a step-by-step worked example

This document follows one real dataset through the whole Market Sonata engine: **NVIDIA (NVDA) daily prices from 2024-01-02 to 2024-12-31**, 252 trading sessions from Yahoo Finance. Every number, chart and note below was produced by running the project's code on that data, so you can reproduce it with `python demo.py NVDA 2024-01-01 2024-12-31`.

The engine is split into three layers, mirroring how a human composer works. The **analysis** layer listens to the market and describes it in a small vocabulary (momentum, volatility, volume, trend). The **composition** layer turns that description into musical decisions (key, tempo, a theme and its development, harmony, dynamics). The **notation** layer writes the result as a MIDI file.

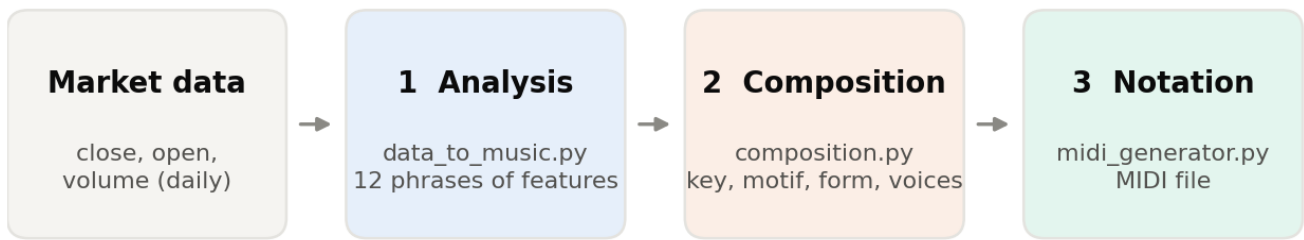


Figure 1. The three-layer pipeline. The composition layer never sees a raw price.

The key design decision

The obvious approach is *one trading day = one note*. It produces 252 notes of jittery, directionless sound: the ear cannot hear a shape in it. Market Sonata instead cuts the year into **12 phrases** (about 21 sessions, roughly one month each) and summarises each phrase with a handful of features. A sonata is built from phrases, not from individual samples; this gives the music room to breathe, repeat and develop.

The result at a glance

Market fact	NVDA 2024	Musical consequence
Total return	+178.8%	Overall trend score +0.65: major key
Daily volatility (score)	0.87 of 1	Tempo 102 BPM (range 60-108)
Ticker	NVDA	Home note D# , fixed for this ticker
Max drawdown	-27.0%	Reported by the analyst note
Opening gaps detected	3	Sforzando accents at velocity 127
Output	188 notes	102 beats, about 60 seconds

Step 0. The input data

NVDA had an extraordinary 2024: it rose from \$48.17 to \$134.29 (+178.8%), with a sharp spring rally, a summer sell-off and a choppy autumn. The engine reads three daily series: the **close**, the **open** and the **volume**. Prices are Yahoo's split-adjusted closes, so the 10-for-1 split in June 2024 does not show up as a fake crash.



Figure 2. Close and volume, with the 12 phrase windows (P1-P12). Shading shows the musical form imposed on top: A (exposition), B (development), A' (recapitulation). Triangles mark the opening gaps the engine turns into accents.

Step 1. Analysis: describe the market in musical terms

`extract_features()` in `data_to_music.py` computes two kinds of description. **Whole-piece** features set the global character; **per-phrase** features drive what happens moment to moment. Every feature is squashed into a fixed range (tanh or min-max scaling) so the composer works with the same numbers whatever the stock or price level.

Whole-piece features

Feature	How it is computed	NVDA 2024
trend [-1, 1]	60% tanh(3 x total return) + 40% share of up days (rescaled)	+0.646
avg_volatility [0, 1]	tanh(40 x std of daily returns)	0.867
avg_volume [0, 1]	mean volume / max volume	0.330
max_drawdown [0, 1]	worst peak-to-trough fall of the close	0.270

Per-phrase features

For each of the 12 windows the engine measures: the **contour** (the price shape inside the window, normalised to [-1, 1] and resampled to 8 points); **momentum** (net change across the window); **volatility** (the window's daily-return spread relative to the whole year); **volume** (mean volume relative to the year's peak); and **gap** (the largest overnight jump, if any).

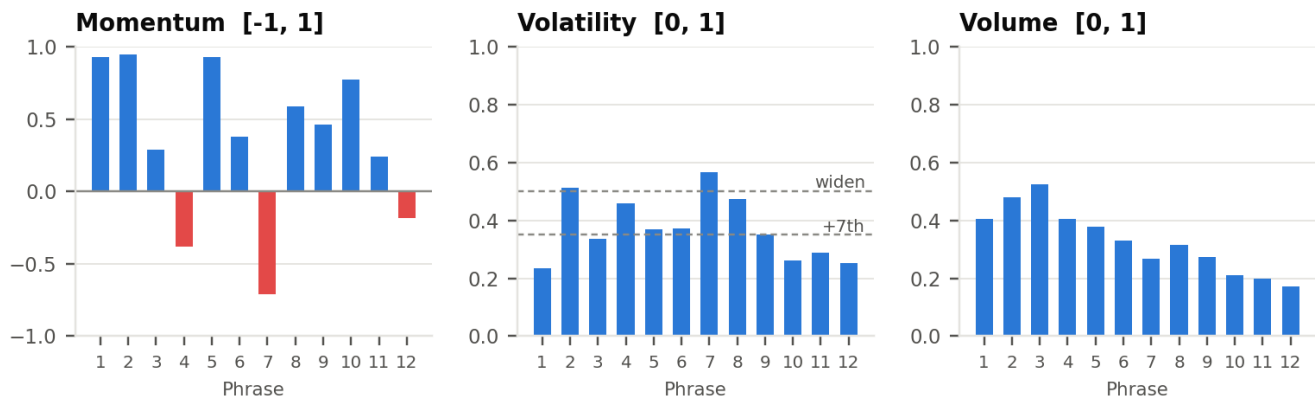


Figure 3. The three per-phrase features that drive the composer. Red momentum bars are falling phrases (P4, P7, P12). Dashed lines are the volatility thresholds used later: above 0.35 the chord gains a 7th; above 0.5 the motif's intervals are widened.

Step 2. Composition: the global decisions

The Composer class first fixes three things for the whole piece. Each is a direct, explainable consequence of a whole-piece feature.

Decision	Rule	NVDA 2024 result
Mode	trend > 0.15: major. trend < -0.15: minor. Otherwise Dorian (neither happy nor sad).	trend +0.65 gives major
Key (home note)	SHA-256 hash of the ticker, mod 12. Every ticker always sings from the same note.	D# (D#4)
Tempo	60 + 48 x avg_volatility. Calm markets breathe slowly; volatile ones race.	60 + 48 x 0.87 = 102 BPM
Chord progression	Major: I-V, V-vi, I, vi-ii, repeating every 4 phrases.	I-V / V-vi / I / vi-ii

Step 3. The motif: one theme for the whole piece

A random-sounding melody is forgettable. So the engine derives a single four-note **motif** from the first phrase, then reuses it everywhere, transformed. The contour of January's price (P1) is mapped onto scale degrees 0-4, and four evenly spaced points are kept: start, rise, peak, resolve.

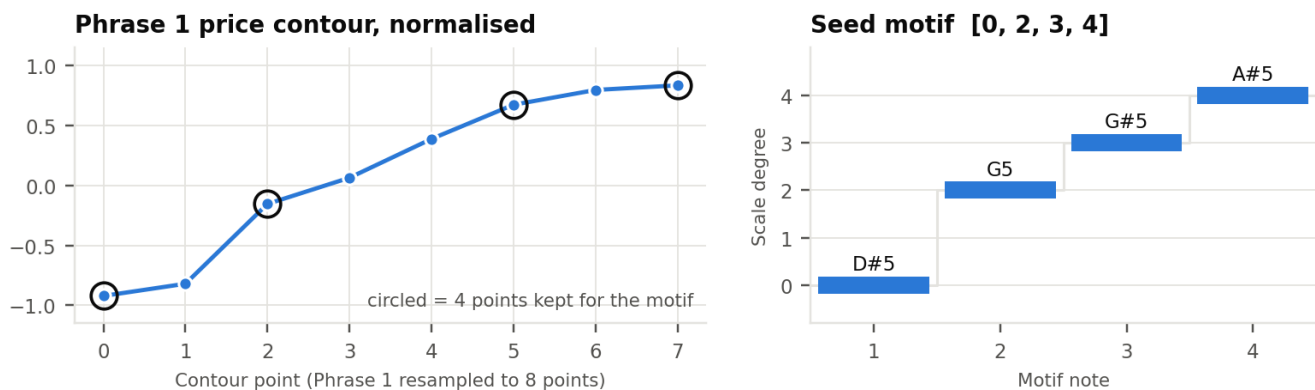


Figure 4. January's steady climb becomes a rising four-note motif, scale degrees [0, 2, 3, 4]. Played one octave above the key's home register these are D#5, G5, G#5, A#5.

Step 4. Form and thematic development

The 12 phrases are grouped into an **A - B - A'** arch, the backbone of classical sonata form. In **A** (P1-P4) the motif is stated plainly. In **B** (P5-P8) and **A'** (P9-P12) each phrase transforms the motif according to what the market did:

Market condition	Transformation	What the ear hears
momentum < -0.2 (falling)	Inversion : flip the motif upside-down around its first note	The theme turns downward
strong momentum	Transposition by round(2 x momentum) scale steps	The theme climbs or sinks
volatility > 0.5	Widening : every interval x 1.5	Bigger leaps, more strain

The table below is the full composition plan for NVDA 2024. Read across a row to see how one month of the market becomes one eight-beat phrase of music.

#	Dates	Market character	Sec.	Motif transform	Motif	Mel. notes	Bass notes	Chord	Vel.
P1	01-02 to 01-31	calm, rising mom +0.93, vol 0.23	A	as stated	[0, 2, 3, 4]	3	4	I-V triad	69
P2	02-01 to 03-01	restless, rising mom +0.95, vol 0.51	A	as stated	[0, 2, 3, 4]	5	5	V-vi +7th	74
P3	03-04 to 04-02	restless, rising mom +0.29, vol 0.34	A	as stated	[0, 2, 3, 4]	4	5	I triad	77
P4	04-03 to 05-01	restless, falling mom -0.38, vol 0.46	A	as stated	[0, 2, 3, 4]	5	5	vi-ii +7th	69
P5	05-02 to 05-31	restless, rising mom +0.93, vol 0.37	B	shift +2	[2, 4, 5, 6]	4	4	I-V +7th	67
P6	06-03 to 07-02	restless, rising mom +0.38, vol 0.37	B	shift +1	[1, 3, 4, 5]	4	4	V-vi +7th	63
P7	07-03 to 08-01	restless, falling mom -0.71, vol 0.57	B	invert, shift -1, widen x1.5	[-1, -4, -6, -7]	5	4	I +7th	59
P8	08-02 to 08-30	restless, rising mom +0.59, vol 0.47	B	shift +1	[1, 3, 4, 5]	5	4	vi-ii +7th	62
P9	09-03 to 10-01	restless, rising mom +0.46, vol 0.35	A'	shift +1	[1, 3, 4, 5]	4	4	I-V triad	59
P10	10-02 to 10-30	calm, rising mom +0.78, vol 0.26	A'	shift +2	[2, 4, 5, 6]	4	3	V-vi triad	55
P11	10-31 to 11-29	calm, rising mom +0.24, vol 0.29	A'	as stated	[0, 2, 3, 4]	4	3	I triad	54
P12	12-02 to 12-31	calm, falling mom -0.19, vol 0.25	A'	as stated	[0, 2, 3, 4]	4	3	vi-ii triad	52

Dates are month-day in 2024. **Mel. notes**: more volatile phrases get more, shorter notes (2 to 8 per phrase). **Bass notes**: density of the left-hand ostinato, driven 60% by volume and 40% by volatility. **Chord**: roots from the progression, coloured by volatility. **Vel.**: MIDI loudness from volume (40 to 112).

Example: P7 (07-03 to 08-01). NVDA fell hard in July (momentum -0.71) and swung widely (volatility 0.57). All three transformations fire: the motif [0, 2, 3, 4] is inverted to [0, -2, -3, -4], shifted down one step to [-1, -3, -4, -5], then widened to [-1, -4, -6, -7]. The cheerful rising theme from January becomes a steep, falling line, and this is where the development section reaches its darkest point.

Step 5. Shocks: opening gaps become accents

A **gap** is when a stock opens far from the previous day's close, usually because news arrived while the market was shut (earnings, a macro shock). No trades happened in between, so the price literally skips. The engine computes $\text{open} / \text{previous close} - 1$ for every session and keeps only jumps larger than 2.5x the typical daily move (and at least 1%), so accents stay rare enough to be heard as events.

Session	Gap	What happened	Musical accent
2024-02-22	+11.2%	Blow-out Q4 earnings released the evening before	High tonic + 3rd + 5th, two octaves up
2024-07-31	+8.8%	Chip stocks rallied after AMD's strong AI results the evening before	High tonic + 3rd + 5th, two octaves up
2024-08-05	-14.2%	Global sell-off as the yen carry trade unwound	Low tonic + 5th, three octaves down

Each accent lasts half a beat at velocity $100 + 400 \times |\text{gap}|$ (capped at 127), placed at the point inside the phrase where the gap occurred. Upward gaps flash in the melody's high register; downward gaps thud in the deep bass.

Step 6. The finished score

Each phrase is rendered in three voices, like a pianist's two hands plus sustain: the **melody** plays the (transformed) motif on the phrase's rhythm, nudged by the price contour; the **harmony** holds the phrase's chords in turn (a two-chord step such as I-V gives each chord half the phrase); the **bass** runs a minimalist broken-chord ostinato in the style of Philip Glass, following whichever chord is sounding. After the last phrase, a final tonic chord across three registers brings the piece home.

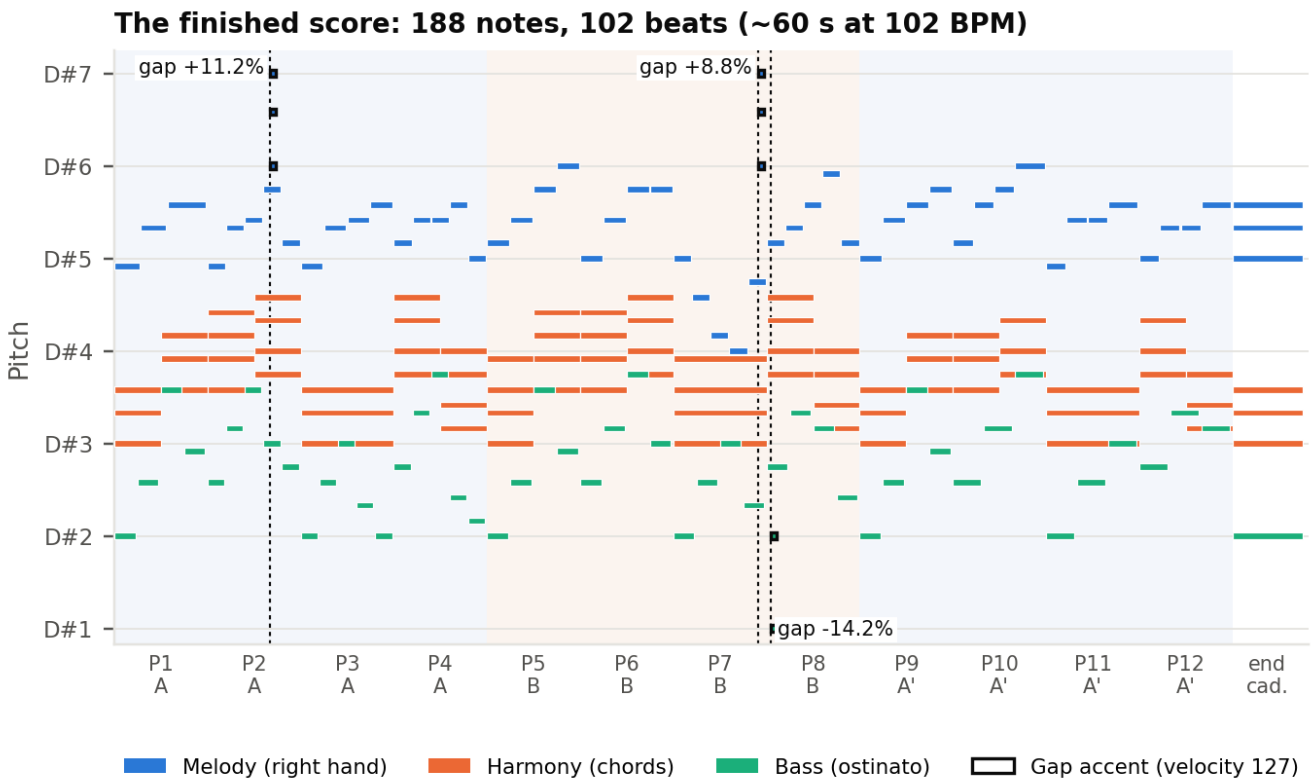


Figure 5. Piano-roll view of the generated MIDI. Time runs left to right in beats (8 beats per phrase); height is pitch. Note how the melody drops in P7, how bass and melody thin out in the quieter A' section as volume fades, and the outlined accents at the three gaps.

Reading the music against the market

- **A (Jan-Apr):** the rising motif is stated four times over a bright major progression. The Q4-earnings gap in February lands as a high accent in P2.
- **B (May-Aug):** the theme is pushed up during the May-June rally, then inverted and stretched in July as the stock sold off, with the early-August crash marked by a low thud at the start of P8.
- **A' (Sep-Dec):** momentum calms and volume fades, so the theme returns closer to its original form, played more softly (velocity falls from 59 to 52), and the piece resolves on the tonic.

Appendix. A general recipe: any time series to a melody

Market Sonata's phrase-and-motif design is one answer. If you just need a single melodic line from any column of numbers, these nine corrections turn raw data into something singable. Each fixes a specific way raw data sounds wrong.

#	Step	Problem it fixes	How
1	Clean	Missing values	Linear interpolation
2	Detrend	A rising series climbs off the keyboard	Subtract the start-to-end line
3	Smooth	Day-to-day noise makes the line jitter	Moving average
4	Downsample	Too many notes to hear a shape	Interpolate to N points (here 16)
5	Normalise	Values must map to a pitch range	5th-95th percentile to [0, 1], robust to outliers
6	Quantise to a scale	Out-of-key 'wrong' notes	Round to scale degrees (major / minor)
7	Limit leaps	Unsingable jumps	Clamp each step to at most 4 degrees
8	End on the tonic	The line never feels finished	Snap the last note to the home note
9	Merge repeats	Mechanical repeated notes	Join equal neighbours into longer notes

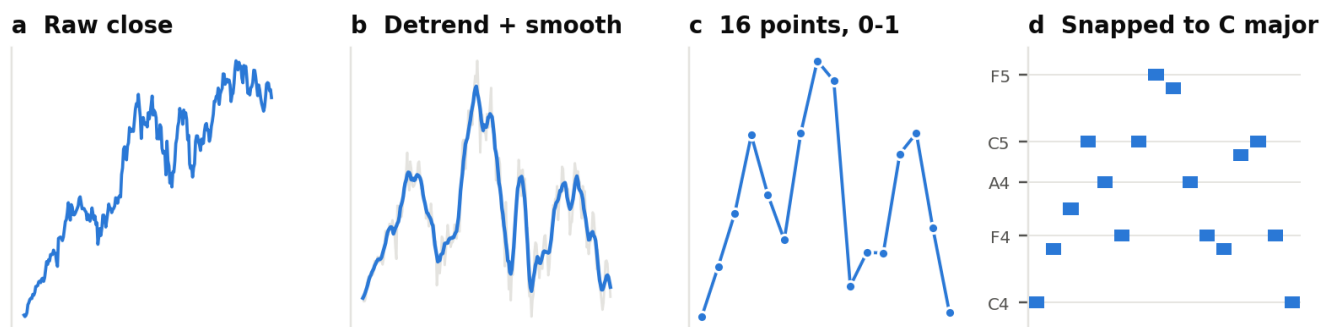


Figure 6. The recipe applied to the same NVDA closes. Resulting 16-note melody in C major: C4 E4 G4 C5 A4 F4 C5 F5 E5 A4 F4 E4 B4 C5 F4 C4.

Core code

```
deg = np.round(normalised * span).astype(int)
for i in range(1, len(deg)):
    deg[i] = np.clip(deg[i], deg[i-1] - max_leap, deg[i-1] + max_leap)
deg[-1] = round(deg[-1] / 7) * 7
pitch = root + 12 * (d // 7) + MAJOR[d % 7]
```

6. quantise
7. limit leaps
8. end on tonic
degree -> MIDI

Design choice worth making deliberately: detrending removes the long-run direction from the melody. If that direction matters (a bull versus a bear year), express it through another musical dimension instead, as Market Sonata does by choosing a major or minor key from the trend.